Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

Winter Semester 2025 – 2026

BCSE309L Cryptography and Network Security

CASE STUDY ASSIGNMENT

SLOT :B1+TB1

Question

Design of a Context-Specific security mechanism Solution for Real-World problem

Identify any **real-world problems.** Example consider the following areas:

- Data Breaches
- Ransomware Attacks
- Phishing and Social Engineering
- Distributed Denial of Service (DDoS)
- Zero-Day Vulnerabilities
- Insider Threats
- Weak Authentication & Password Issues
- Cloud Security Misconfigurations
- IoT Security Vulnerabilities
- Advanced Persistent Threats (APT)
- Supply Chain Attacks
- Data Integrity Attacks

Then, perform the following tasks:

1. **Clearly define the problem statement**, including its real-world context and motivation.
2. **Analyze at least two existing solutions** related to the identified problem. Each existing solution must be supported with **documentary proof**, such as:
   - Research paper / technical article
   - Patent (patent number or title)
   - Commercial or industrial product (product name or official source)

   For each solution:

    - Briefly explain its working
    - Clearly identify its **technical or practical limitations** in the chosen context
3. **Propose a custom security mechanism solution** that addresses the identified limitations.
4. **Discuss feasibility, constraints, and limitations**, including cost, deployment challenges, and scalability.

Important Conditions:

- The proposed solution must be **realistic and implementable** using security mechanism..
- **Block diagram is mandatory**.
- **At least two existing solutions must be supported with proper references**.
- Claims without references will **not be awarded marks**.
- **No requirement to build a software prototype.**

Submission Format (Max 5 Pages)

1. Problem Description (½–1 page)
2. Existing Solutions with Proof & Identified Gaps
3. Proposed System Architecture (with block diagram)
4. Component Selection and Justification
5. Constraints, Cost & Limitations
6. Conclusion
7. References (mandatory)

Each student should give the ppt presentation about your project with maximum of 15 slides on or before 17-04-26

FINAL EVALUATION RUBRIC

Report Evaluation (7 Marks)

S. No	Evaluation Criteria	Excellent	Good / Average	Poor / Inadequate
1	**Problem Definition & Context**	Clearly defined, specific real-world problem with strong contextual relevance (2 Marks)	Problem stated but lacks clarity or context (1 Mark)	Vague, generic, or unrealistic problem (0 Mark)
2	**Existing Solutions with Proof & Gap Analysis**	Two valid existing solutions cited with proper references and clear identification of limitations (2 Marks)	Solutions cited but weak analysis or incomplete references (1 Mark)	No proof, copied, or irrelevant discussion (0 Mark)
3	**Architecture & Block Diagram**	Technically correct architecture with clear block diagram and logical data flow (2 Marks)	Diagram present but lacks technical depth or clarity (1 Mark)	Generic, incorrect, or missing architecture (0 Mark)
4	**Component Selection & Justification**	Appropriate components selected with clear technical justification (cost, power, feasibility)(1 Mark)	Components listed with weak justification (0.5 Mark)	Random components without reasoning (0 Mark)

Individual Viva Voce Evaluation (3 Marks per Student)

S. No	Viva Criteria	Excellent	Good / Average	Poor
1	**Understanding of Assigned Section**	Clearly explains own contribution and design decisions (1.5 Marks)	Partial understanding, minor confusion (0.75 Mark)	Unable to explain (0 Mark)
2	**security Systems Conceptual Knowledge**	Correct and confident application of security concepts (1 Mark)	Basic understanding (0.5 Mark)	Conceptually weak (0 Mark)
3	**Clarity, Confidence & Response**	Clear, logical, and confident responses (0.5 Mark)	Hesitant or unclear (0.25 Mark)	No clarity (0 Mark)